

PRODUCT REQUIREMENTS DOCUMENT

ShopEase

Checkout Optimization for E-Commerce Platform

Author	Pratiksha
Version	v1.0 — Draft
Date	June 2026
Domain	E-Commerce / Checkout UX
Status	V1 Complete

CONFIDENTIAL — PORTFOLIO PROJECT

1. Problem Statement

ShopEase, an e-commerce platform, has experienced a significant increase in cart abandonment over the last 3 months. Customers are successfully adding products to their cart but dropping off just before payment completion. This directly impacts revenue, customer retention, and overall platform growth.

The core challenge is to diagnose and eliminate friction in the checkout flow so that conversion rates improve, customer experience is enhanced, and revenue is recovered.

2. Business Goals

- Reduce cart abandonment rate from 72% to 55%
- Improve checkout conversion rate from 65% to 75%
- Increase successful payment transactions
- Reduce average checkout completion time
- Improve customer experience during checkout
- Increase customer retention rate

3. Key Business Questions

- At what stage of checkout are customers dropping off?
- Is the problem across all product lines or specific categories?
- Is the drop-off happening on mobile app, website, or both? What is the split?
- Is abandonment seasonal or consistent throughout the year?
- Which payment method shows the highest drop-off rate?
- Is the abandonment coming from guest users or logged-in customers?
- Does page load time affect checkout completion?
- Are payment failures contributing significantly to abandonment?

4. Stakeholders

Role	Responsibility
Product Manager	Define product goals, own PRD, drive alignment
Business Analyst	Requirements gathering and analysis
UX Team	Improve checkout experience and wireframes
Developers	Implement solutions
QA Team	Testing and validation
Payment Gateway Team	Payment integration support

Customers

End users — primary stakeholders

5. Scope

In Scope

- Checkout flow analysis
- Payment failure analysis
- Mobile responsiveness analysis
- Guest checkout analysis
- Checkout UX improvements

Out of Scope

- Inventory management
- Product recommendation engine
- Logistics optimization

6. Functional Requirements

- System shall allow guest checkout without account creation
- System shall support multiple payment modes (UPI, Card, Net Banking, Wallet, COD)
- System shall provide a payment retry option on failure
- System shall display a checkout progress indicator
- System shall save cart items for returning users (up to 7 days)

7. Non-Functional Requirements

Requirement	Target
Checkout page load time	≤ 3 seconds
Payment transaction security	PCI-DSS compliant
System traffic support	High concurrency — peak load handling
Mobile responsiveness	Optimized for iOS and Android
API response time (p95)	≤ 1 second

8. Success Metrics

Layer 1 — Product KPIs

Metric	Baseline	Target	Formula
Checkout Conversion Rate	65%	75%	Orders Completed / Checkout Started × 100
Cart Abandonment Rate	72%	55%	Carts Abandoned / Carts Created × 100
Payment Success Rate	50%	> 75%	Successful Transactions / Total Transactions × 100
Payment Failure Rate	16.67%	< 10%	Failed Payments / Total Attempts × 100
Payment Pending Rate	33.33%	< 15%	Pending Transactions / Total Transactions × 100

Layer 2 — Derived Insights (from Prototype Data)

Credit Card Failure Rate: 50% (1 of 2 transactions failed) — possible causes: authentication friction, gateway timeout, bank-side decline, OTP drop-off.

Net Banking Pending Rate: 100% (2 of 2 transactions pending) — possible cause: slower payment confirmation or gateway processing delays.

Although API response times meet NFR thresholds, payment friction (credit card failures, net banking delays) continues to contribute to cart abandonment. Improving payment reliability is a priority for V2.

9. Epics & User Stories

Epic ID	Epic Name	Priority
EP-01	Simplify Checkout Experience	High
EP-02	Reduce Cart Abandonment	High
EP-03	Provide Seamless Payment Options	High
EP-04	Improve Guest Login System	Medium

US-01 — Guest Checkout (EP-01) | Priority: High

As a customer, I want to check-out as a guest so that I can place an order without creating an account.

Acceptance Criteria:

- Given a customer is browsing as a guest, when items are added to the cart, then checkout must complete without mandatory account creation
- Customer must be able to track the order and complete payment as a guest

Business Value: Reduces friction during checkout — primary conversion driver.

US-02 — Progress Indicator (EP-01) | Priority: Medium

As a customer, I want to see checkout progress so that I know how many steps are remaining.

Acceptance Criteria:

- Progress bar must display: Cart → Shipping → Payment → Confirmation
- Remaining steps must be clearly visible at each stage

Business Value: Improves user clarity and reduces drop-off from uncertainty.

US-03 — Multiple Payment Methods (EP-03) | Priority: High

As a customer, I want multiple payment options so that my transaction is seamless.

Acceptance Criteria:

- Supported: UPI, Credit/Debit Card, Net Banking, Wallets, Cash on Delivery
- Transaction must complete within 5 seconds
- Failed transaction must display a clear error with retry option
- Payment data must be secured (PCI-DSS)

Business Value: Improves payment transaction success rate.

US-04 — Save Cart (EP-02) | Priority: Medium

As a customer, I want my cart saved automatically so that I can resume checkout later from any device.

Acceptance Criteria:

- Cart items sync across app and website for logged-in users
- Items saved for up to 7 days
- Abandonment notification sent after 7 days of inactivity
- Out-of-stock or removed items clearly flagged

Business Value: Reduces cart abandonment through recovery mechanism.

10. Proposed Solutions (AS-IS → TO-BE)

Pain Point	Solution	Expected Impact
Forced account creation	Guest Checkout	Reduced friction, higher conversion
Multi-page checkout flow	Single-page checkout	Faster completion, better mobile UX

Manual address entry	Autofill (GPS + Browser)	Reduced effort, faster completion
No coupon visibility	Auto-applied coupons	Higher satisfaction, better retention
Payment failures with no recovery	Payment Retry Mechanism	Higher payment success rate
No order confirmation signal	Real-time notifications (Email + SMS)	Builds trust, improves experience

11. Assumptions & Limitations

- Guest user abandonment tracking is partially simulated in V1
- Current dataset focuses on unsuccessful payments and negative review indicators
- Additional abandonment reasons (high shipping cost, forced registration, slow checkout, delivery concerns) can be incorporated in future iterations
- Payment API integrations (Razorpay, PhonePe, Paytm) are assumed to be available for V2 implementation